DevOps Lab Report

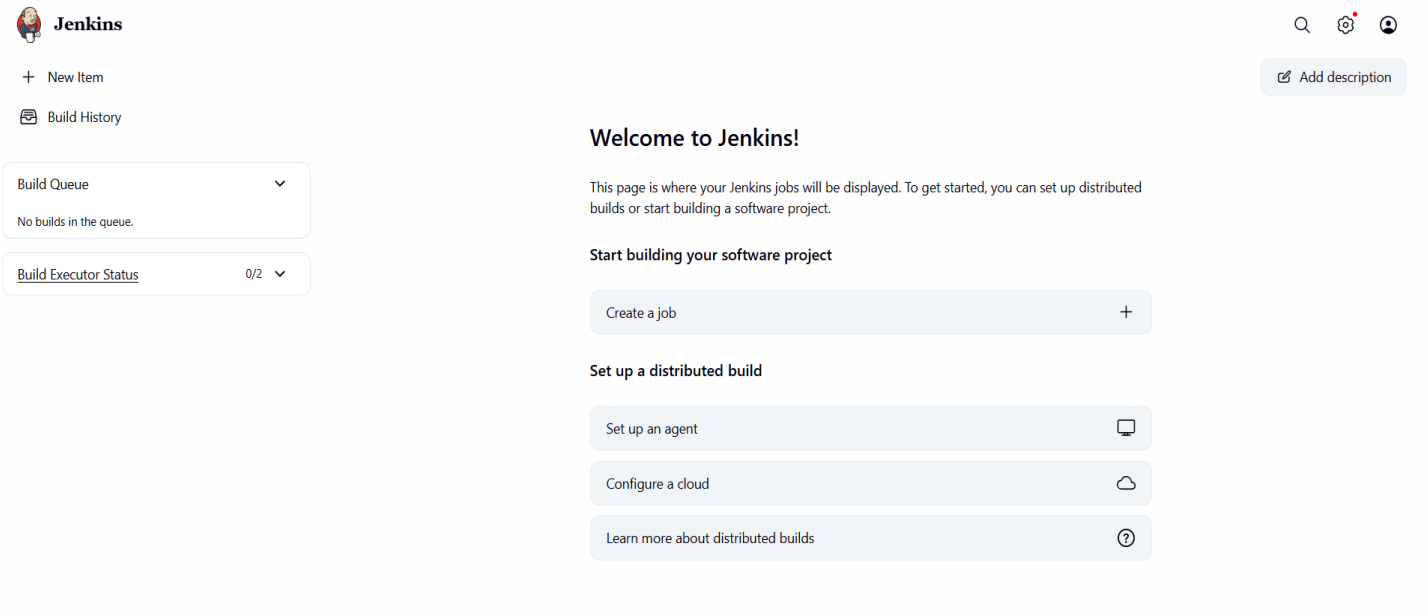
Week – 4

Group Members	USN
Poorvika G J	1MS23IS090
Nikita Bhattad	1MS23IS078

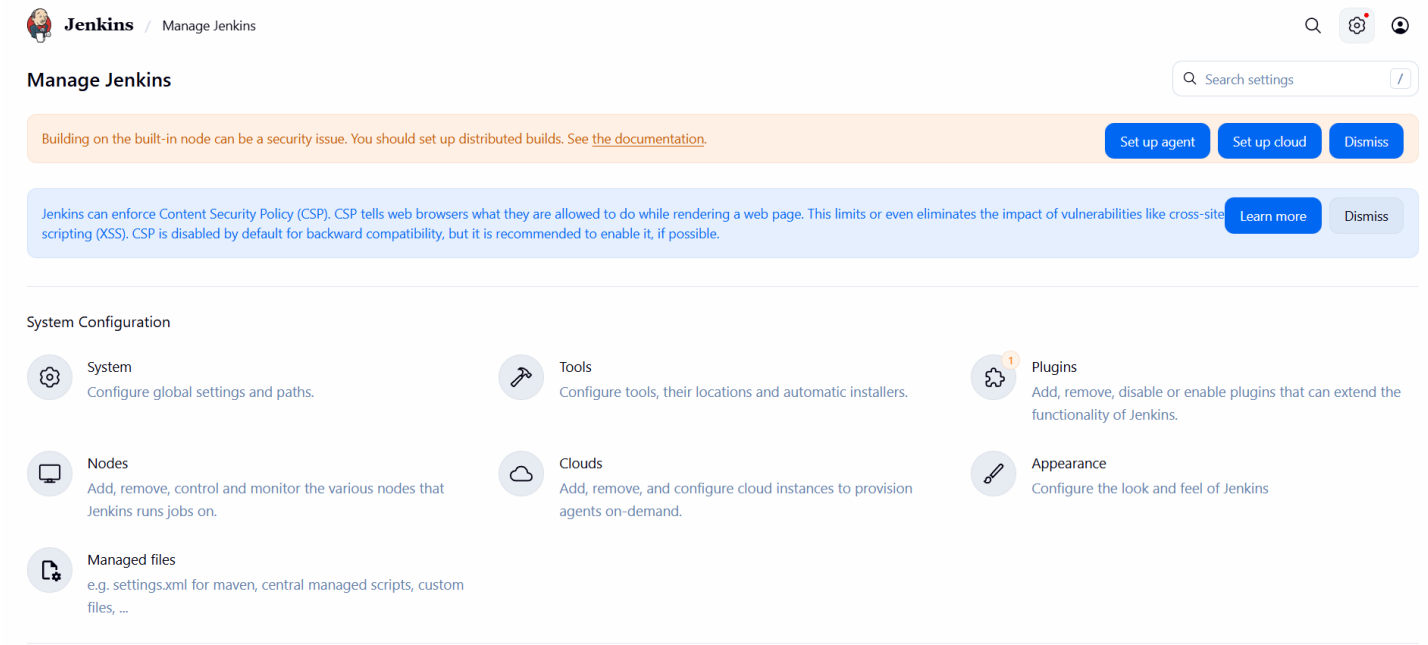
Jenkins Pipeline

Inside the script – write the gravy code

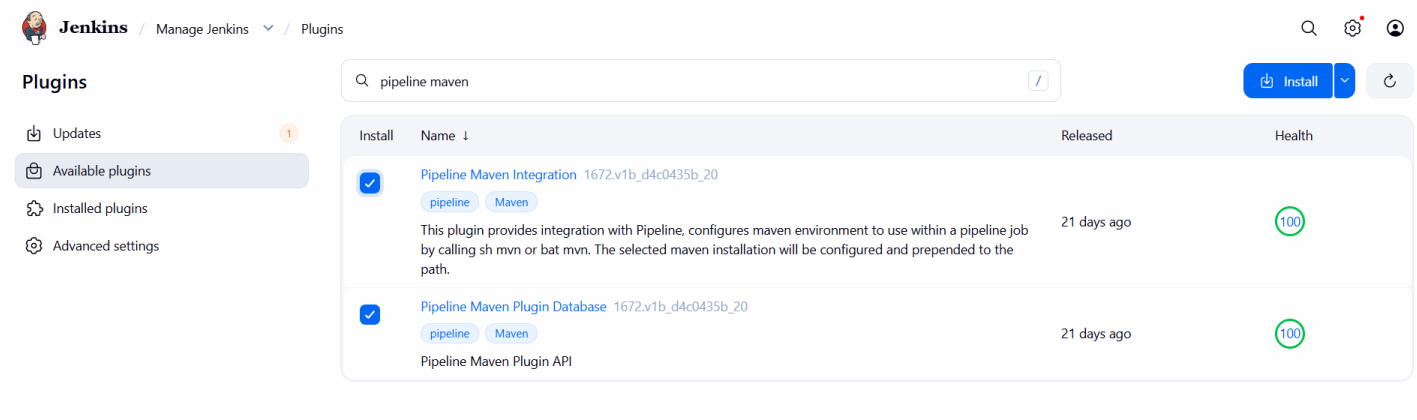
Open Jenkins in the PC which is located in localhost:8080 (default port)



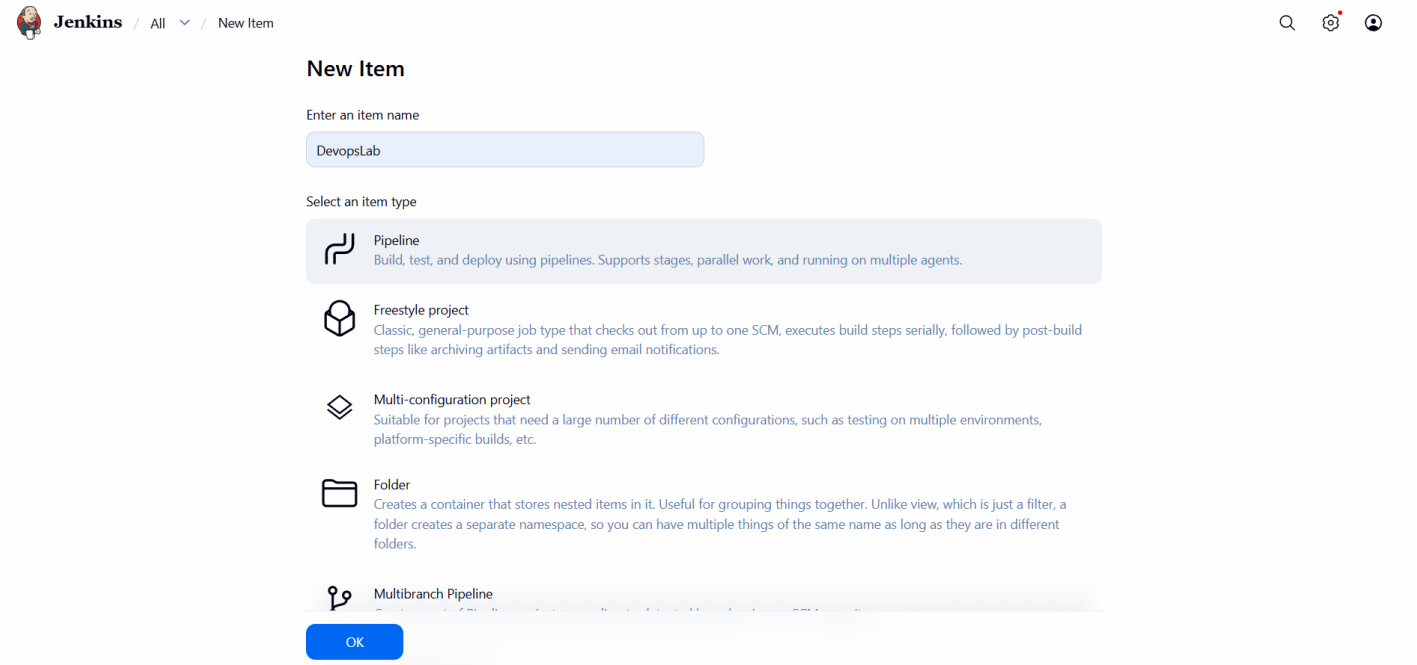
Go to Settings and select plugins



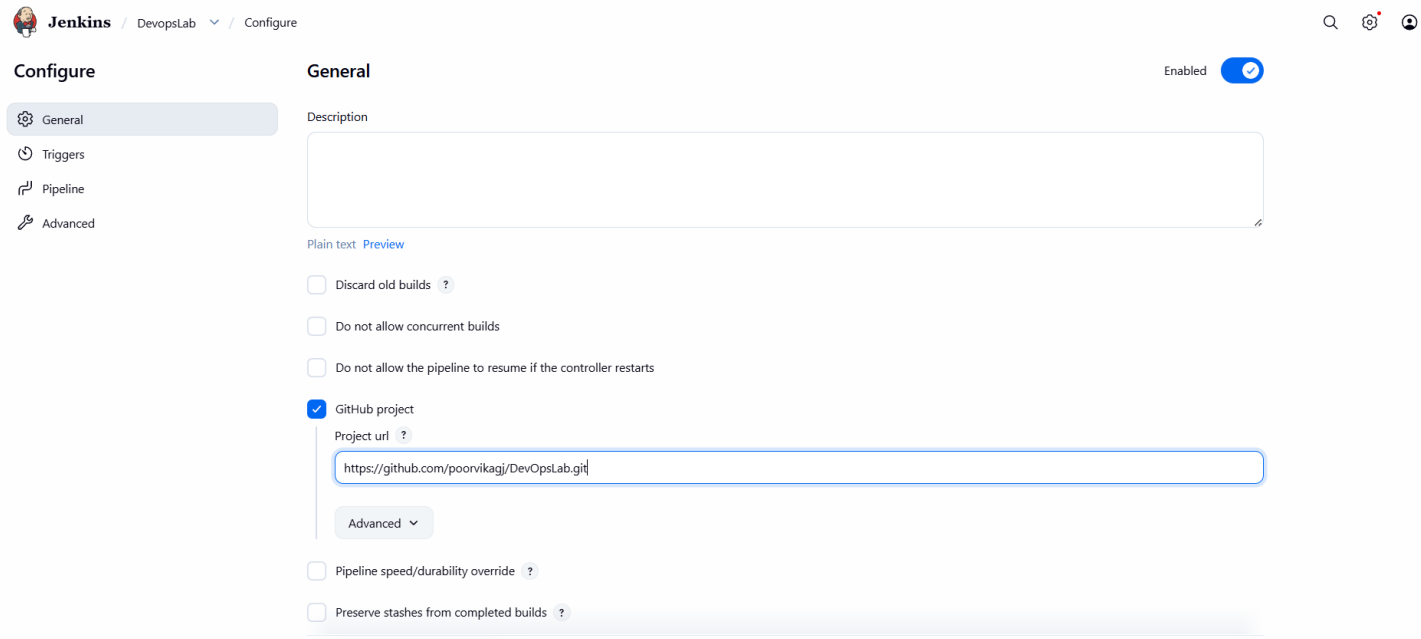
Install **Pipeline Maven Integration, Pipeline Maven Database, Eclipse Temurin installer**



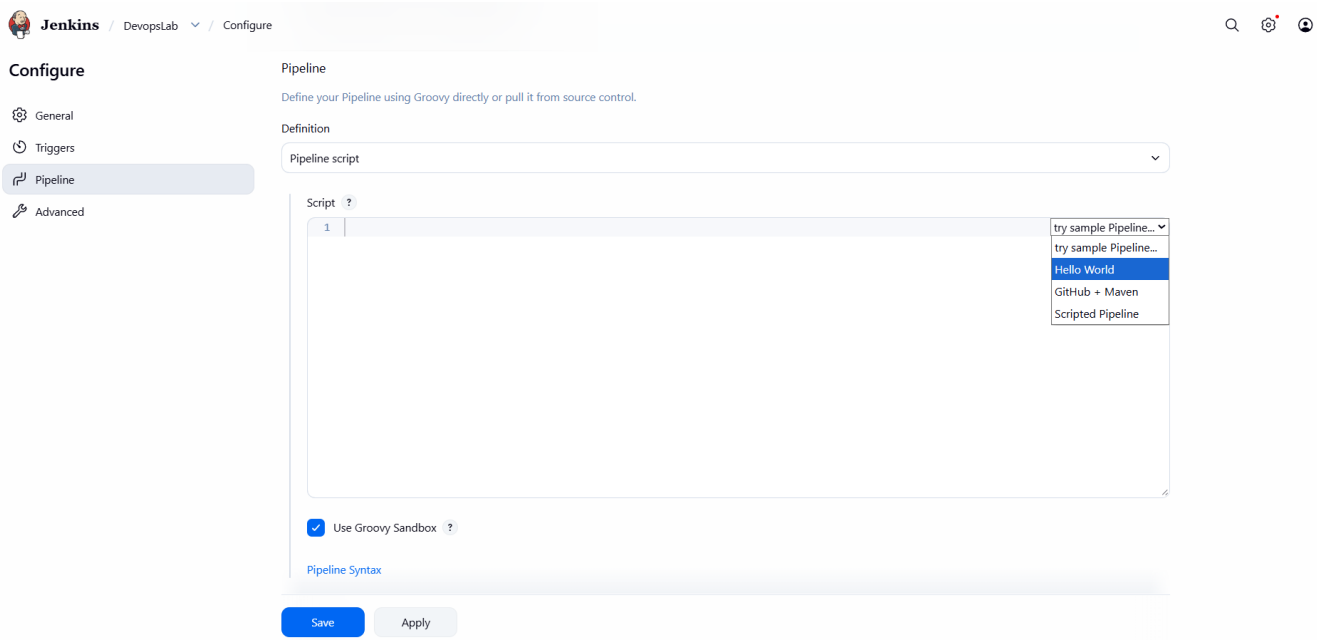
Now go back to the Jenkins home page and click on new item to create a new pipeline
Enter the pipeline name and choose the item type which is pipeline here and click on OK



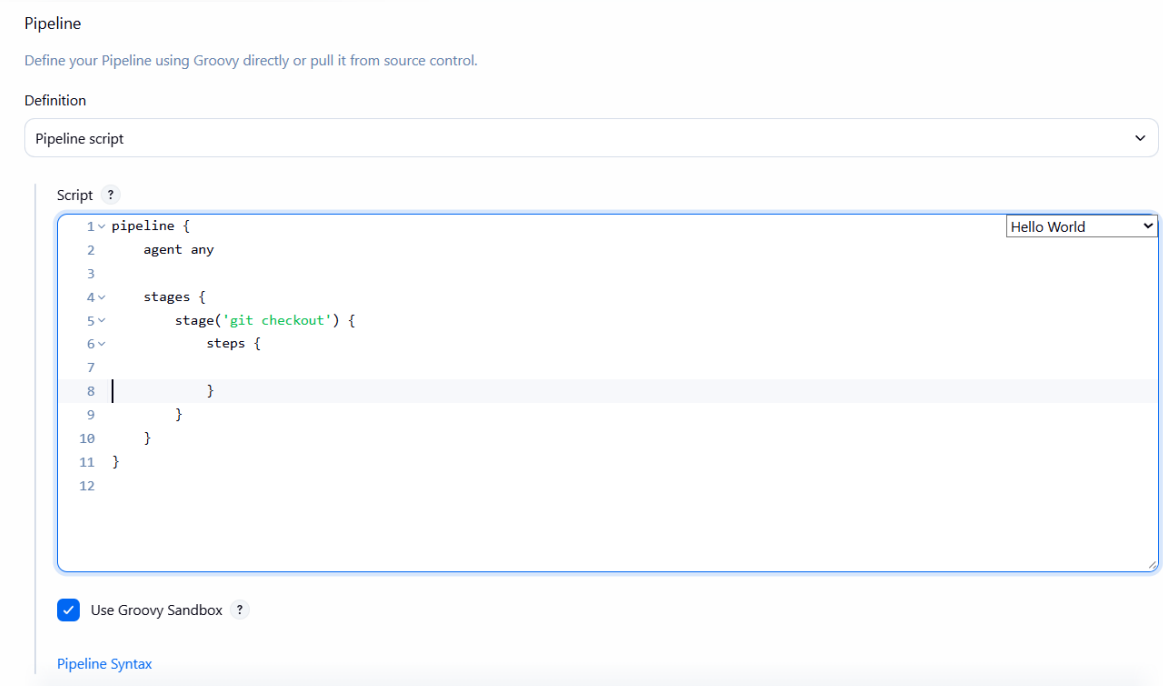
Select Github project as we are using the project present in the github and the respective project github repository link.



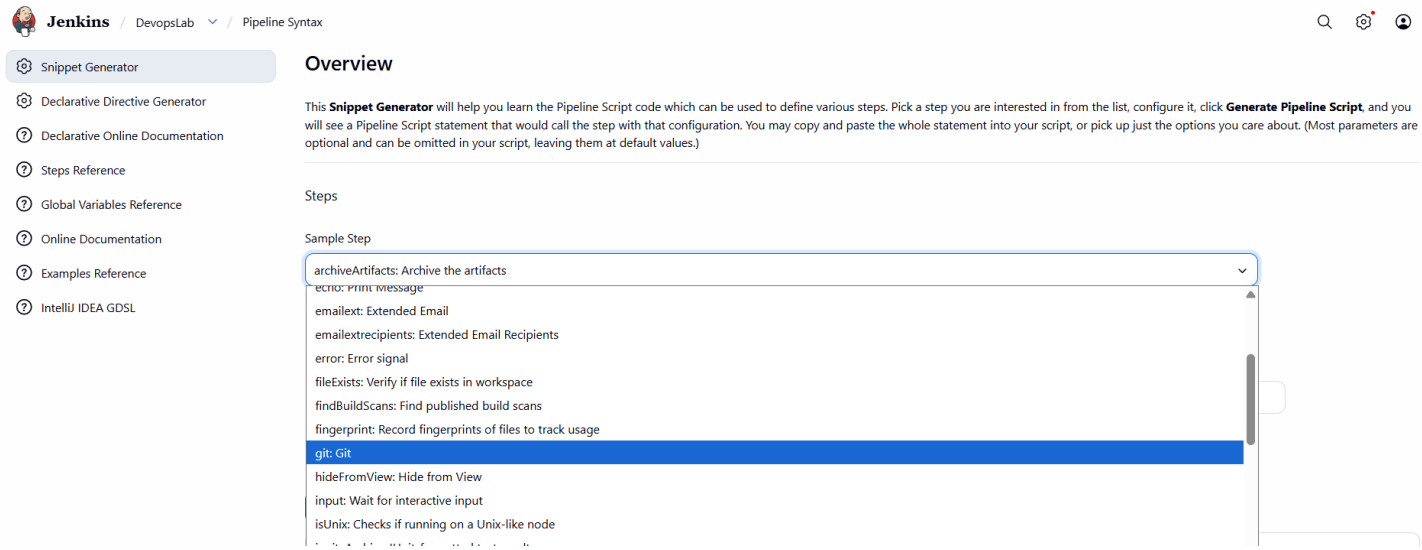
Scrolldown and in the scripts section Select the Hello World (sample grovvy code)



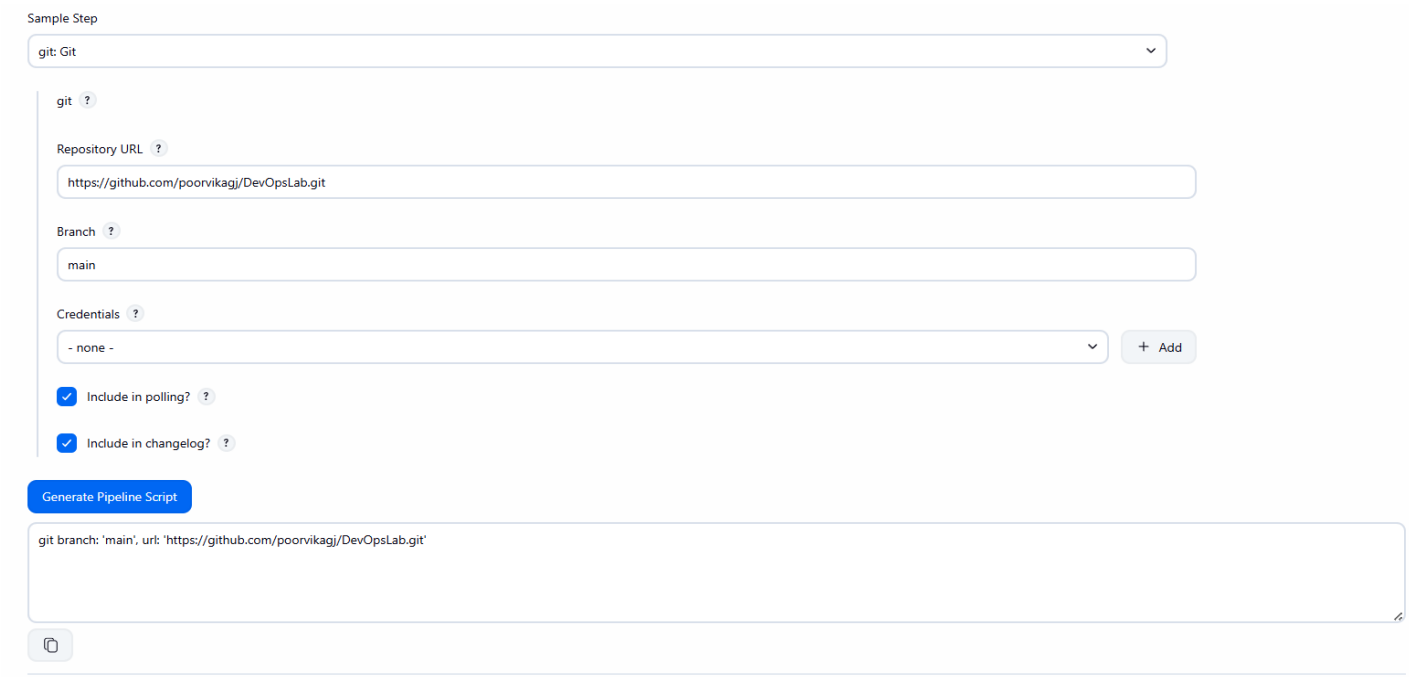
Change the stage name from hello world to git checkout and remove the echo line inside the steps. Then click on the pipeline syntax and it redirects to the respective link.



In the pipeline syntax page select git in sample step as we are using git in the steps.

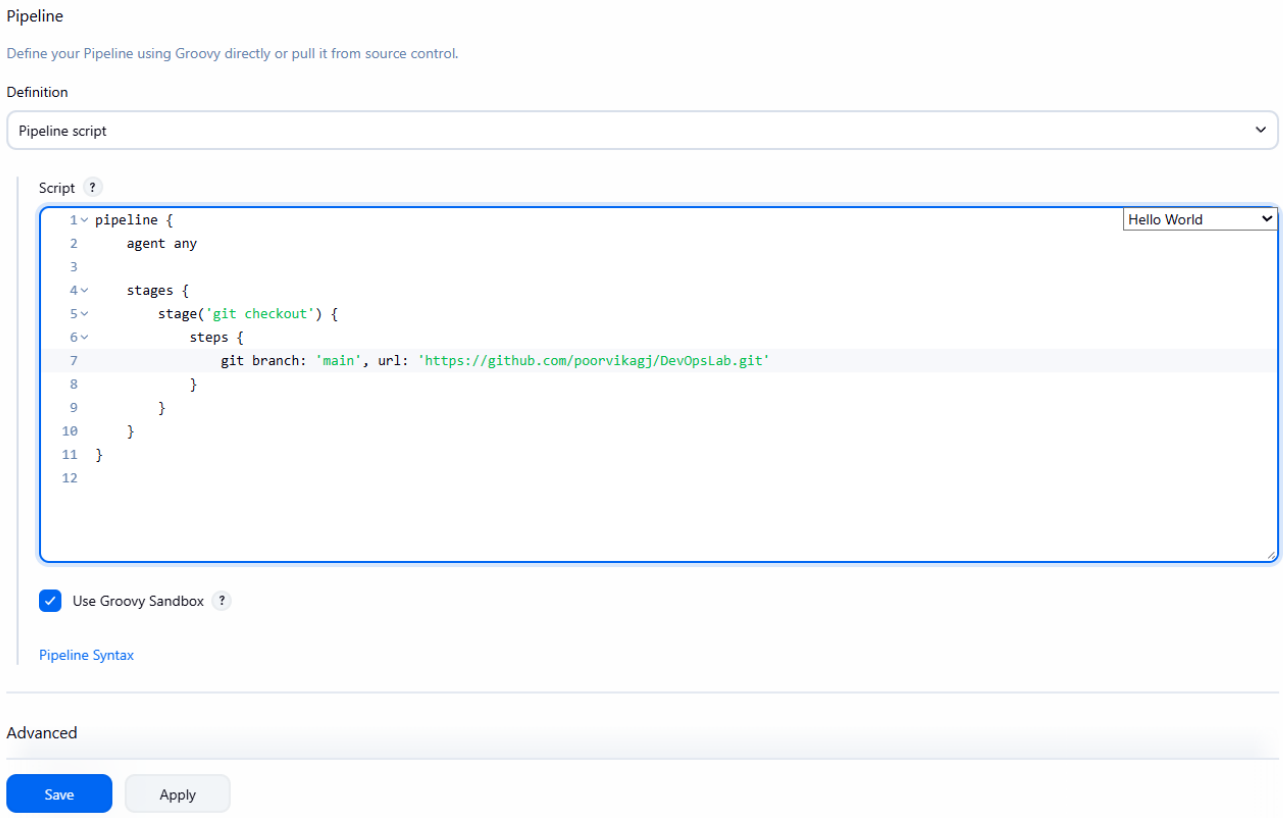


Paste the git repo link and click on generate pipeline script and copy the pipeline script

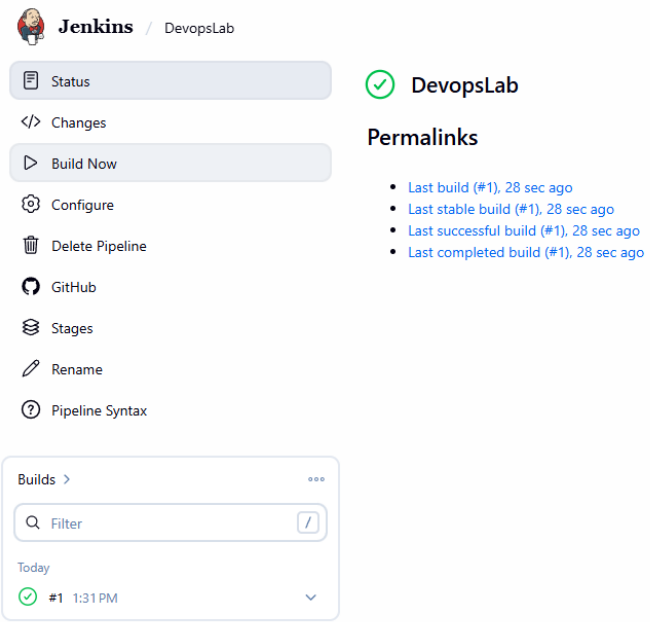


Now come back to the pipeline page and paste the copied pipeline script in the script section where the echo line was removed.

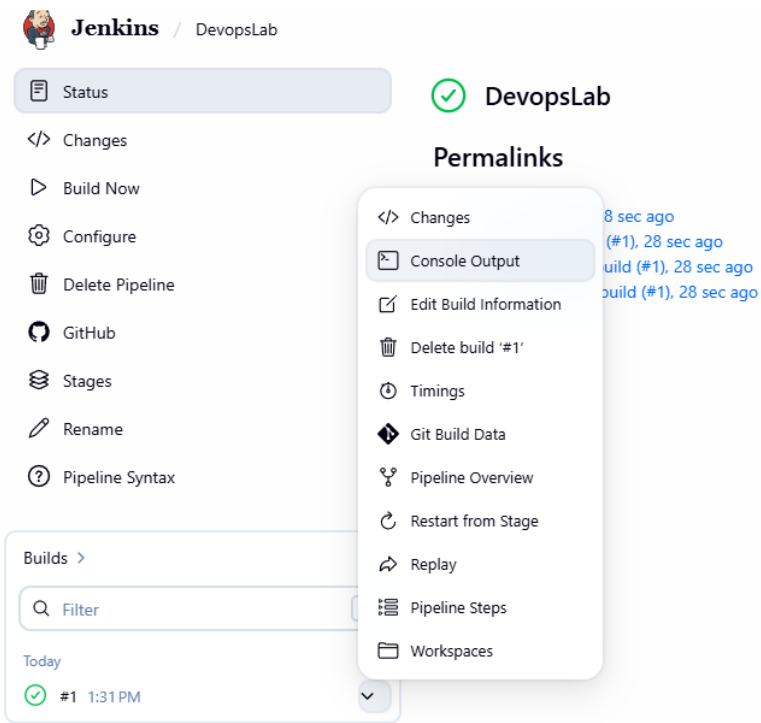
As the stages of this project is completed Click on save.



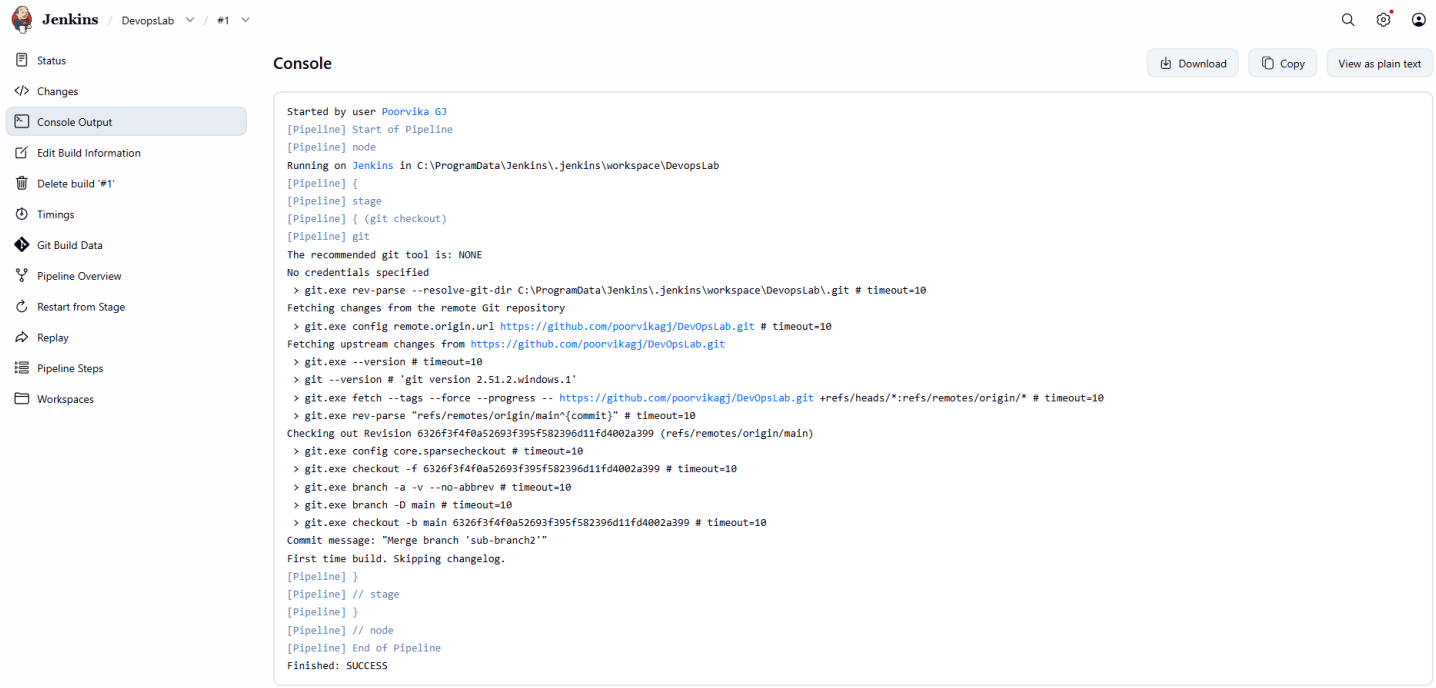
You be inside the saved pipeline and to build the saved pipeline click on Build now.
You can see the last build details.



To check the status or the errors of the pipeline click on build up options button and select the Console Output to check where the error is or if the pipeline buildup is successful.



Console Output



Using the JenkinsFile

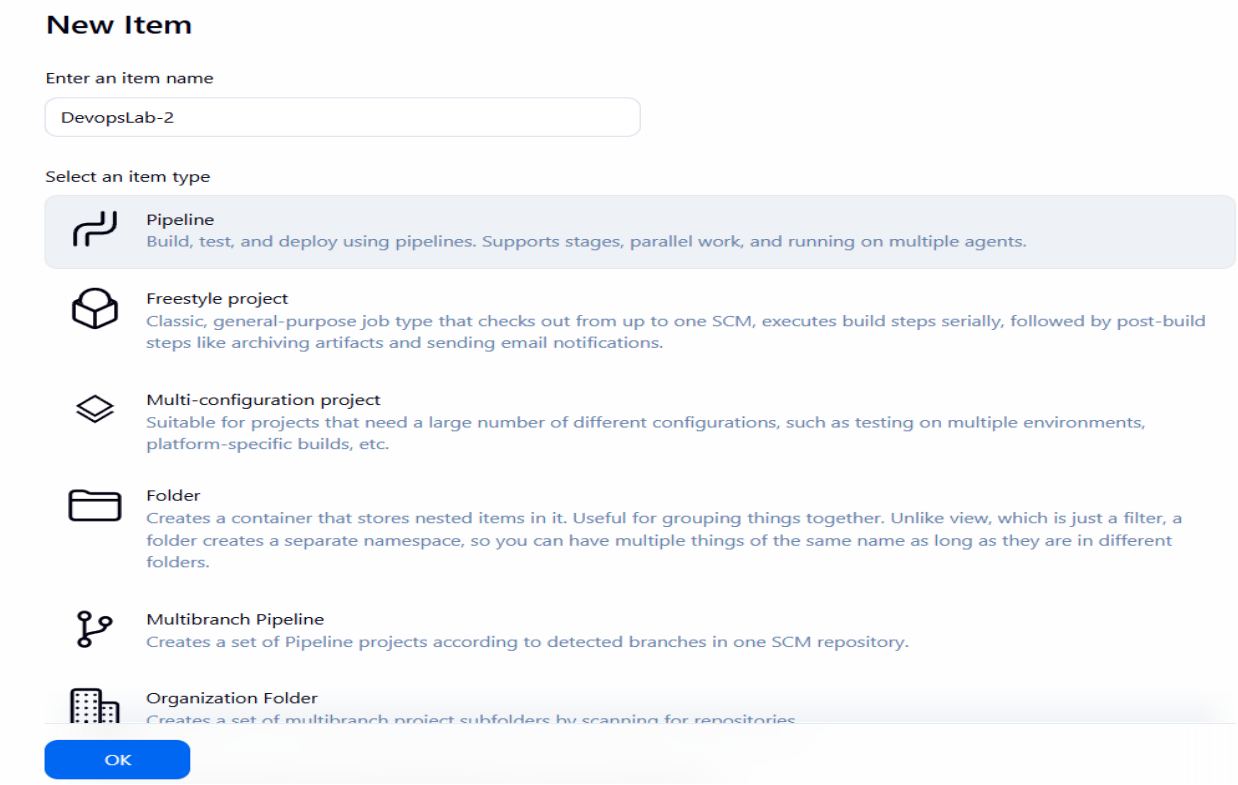
Open the Project you are working and a file named Jenkinsfile with no extension to write groovy code for the pipeline.

After the pipeline setup code. Add the commit the code to the GitHub repository and push the code.

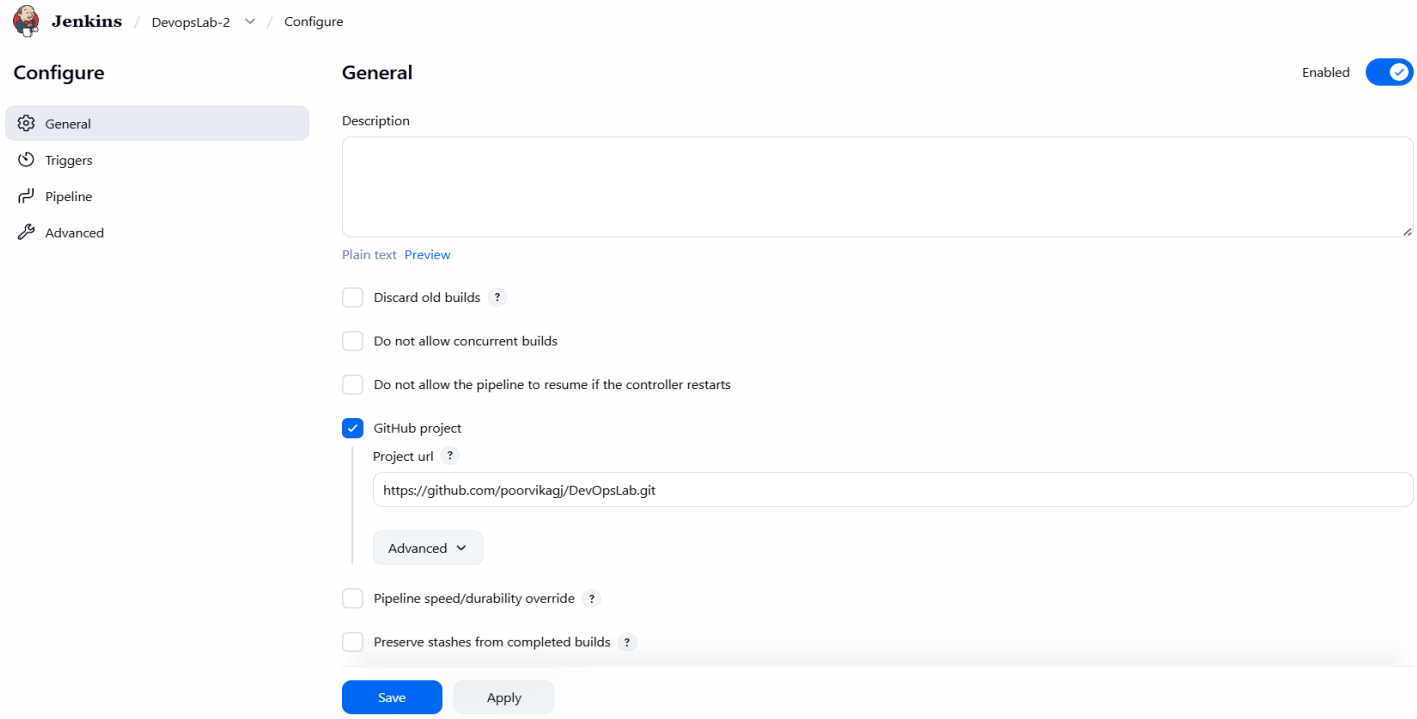
```
HP@DESKTOP-OGSVFEE MINGW64 ~
$ cd devOpsLab/

HP@DESKTOP-OGSVFEE MINGW64 ~/devOpsLab (main)
$ ls
1.txt  2.txt  3.txt  4.txt  Jenkinsfile  README.md

HP@DESKTOP-OGSVFEE MINGW64 ~/devOpsLab (main)
$ git add .

HP@DESKTOP-OGSVFEE MINGW64 ~/devOpsLab (main)
$ git commit -m "Added Jenkinsfile"
[main 999c895] Added Jenkinsfile
 1 file changed, 72 insertions(+)
 create mode 100644 Jenkinsfile
```

```
HP@DESKTOP-OGSVFEE MINGW64 ~/devOpsLab (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 886 bytes | 886.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/poorvikagj/DevOpsLab.git
   6326f3f..999c895  main -> main
branch 'main' set up to track 'origin/main'.
```

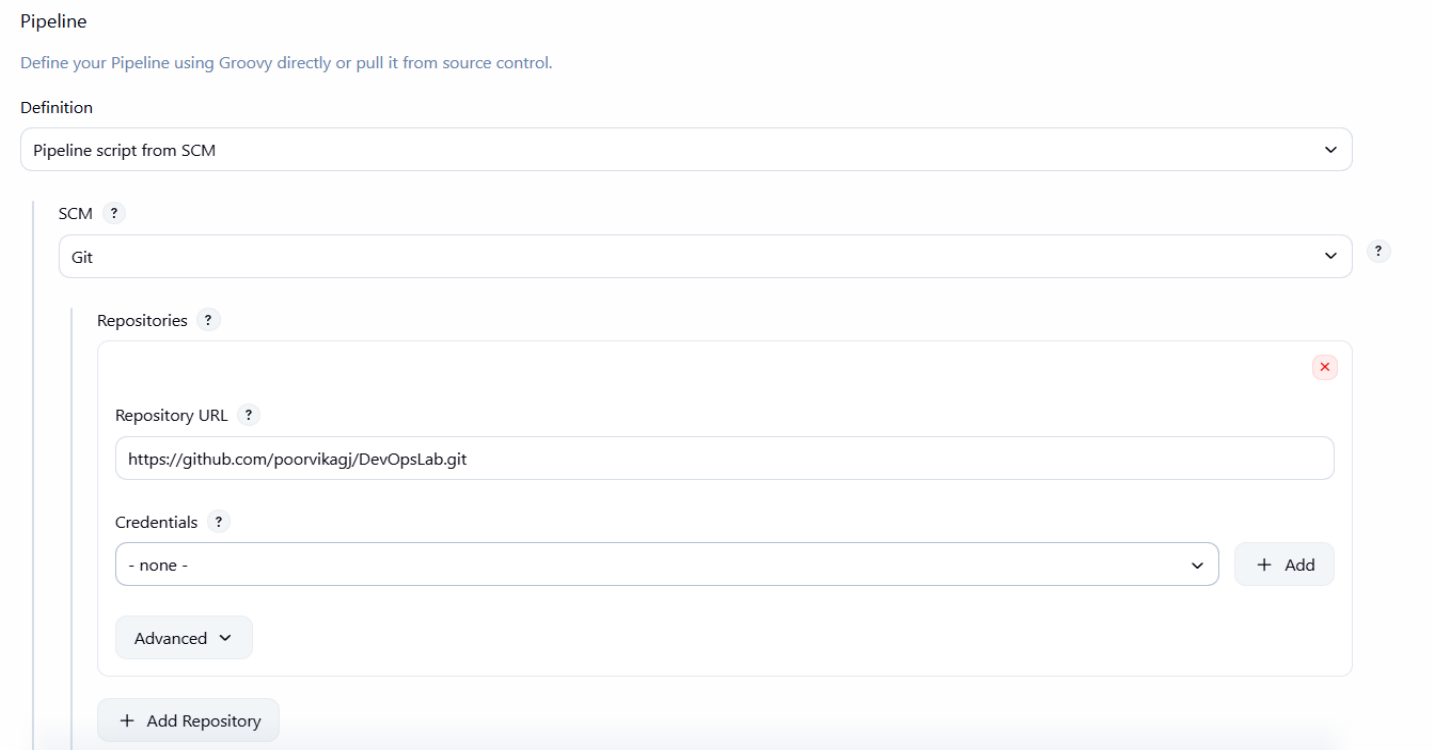
Now go to Jenkins and add a new item to create a pipeline and name it.



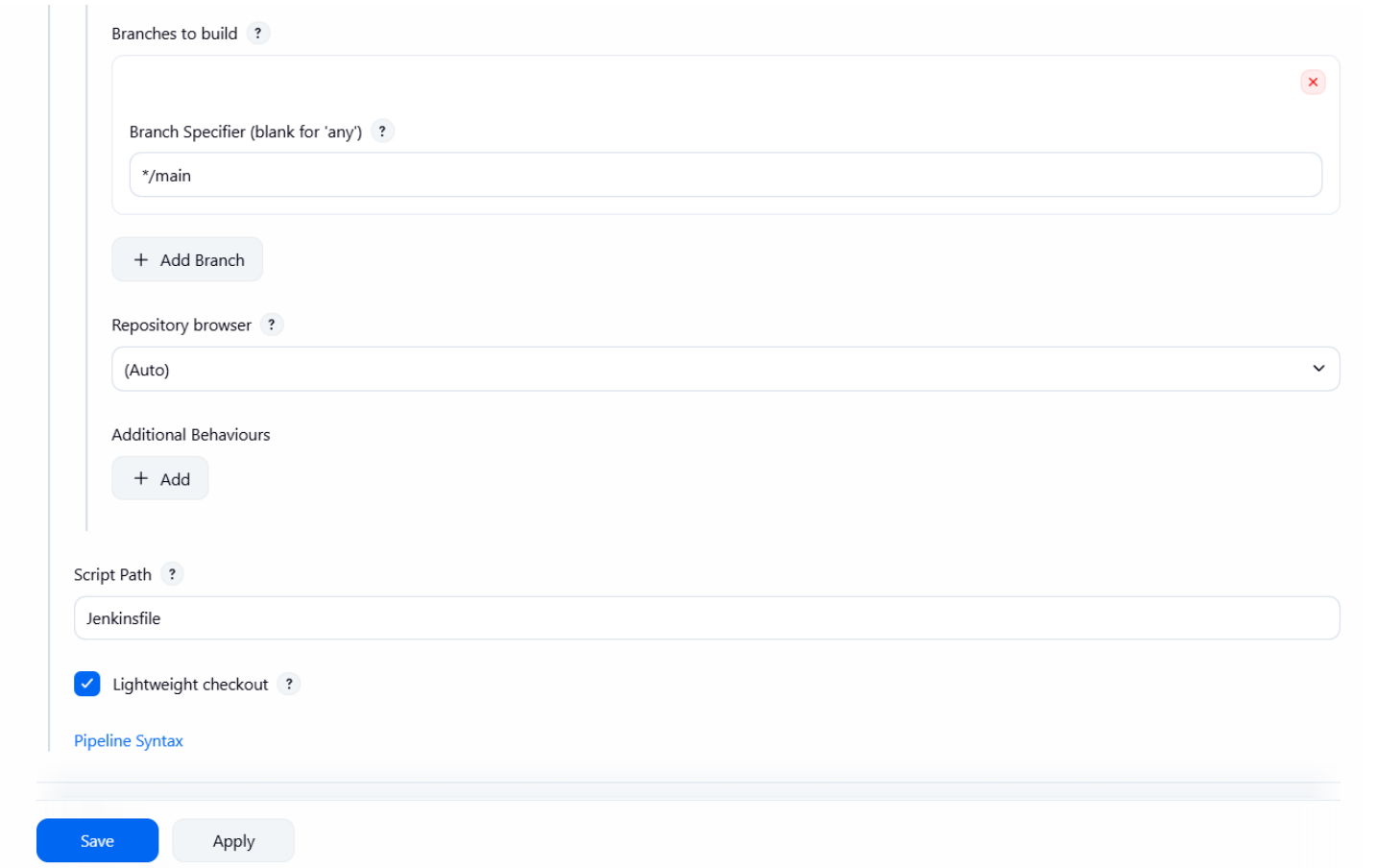
As the project we are setting up is in GitHub select GitHub project and paste the repo url.



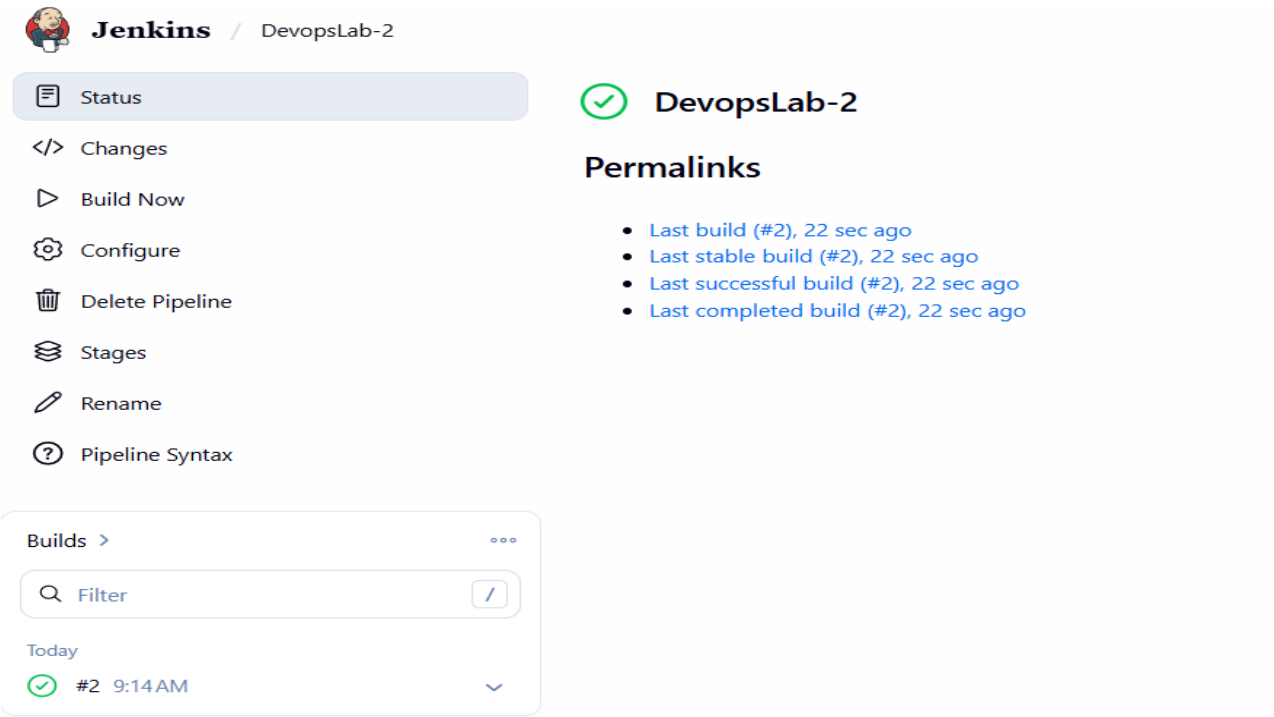
Now the Jenkinsfile is the repo and the pipeline code should be read from the repo- you select pipeline script from SCM and fill in the rest of the details. The Jenkinsfile is autodetected.



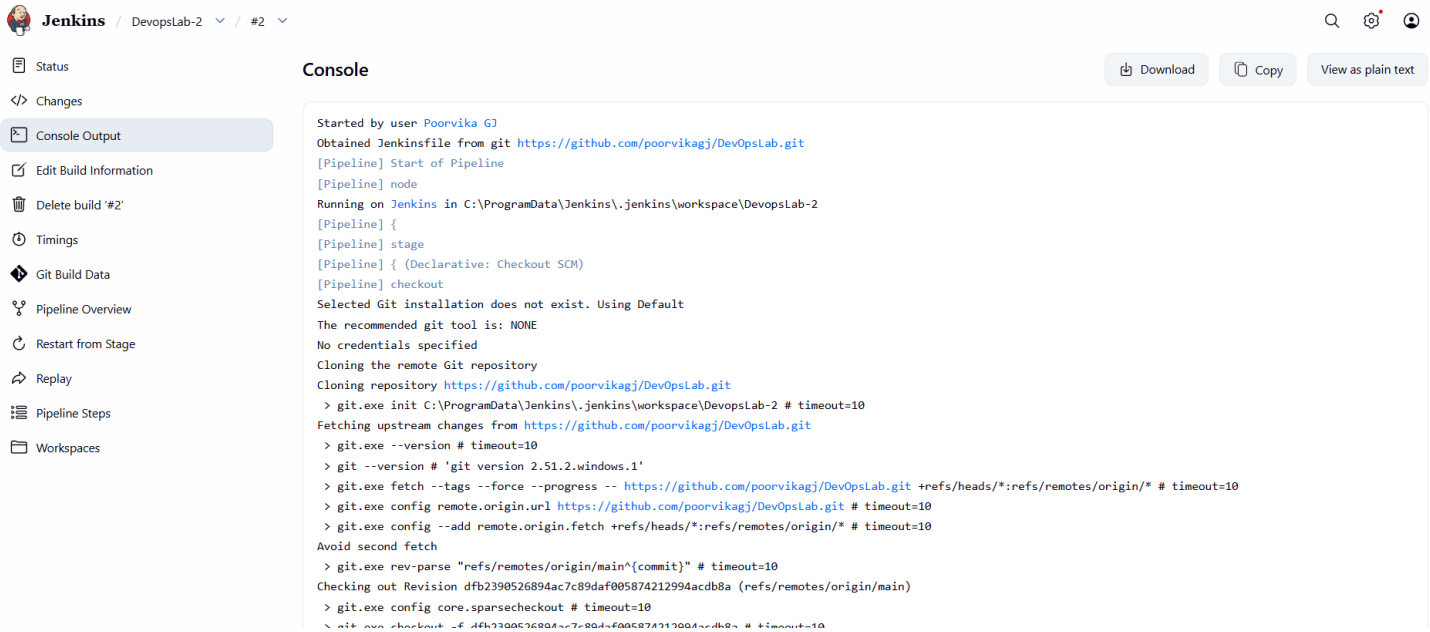
Give details from which branch it should clone and other details and save it.



To build the pipeline click on the build Now option and you can see the building details.

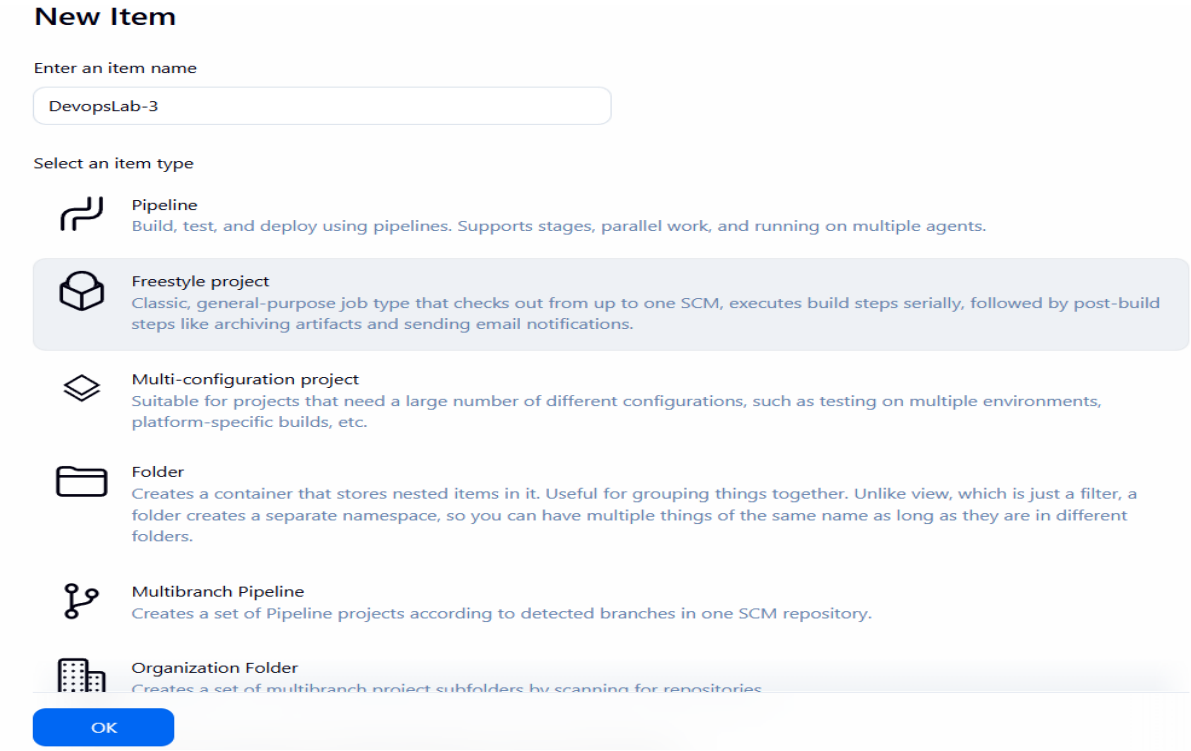


Console output of the build Up.

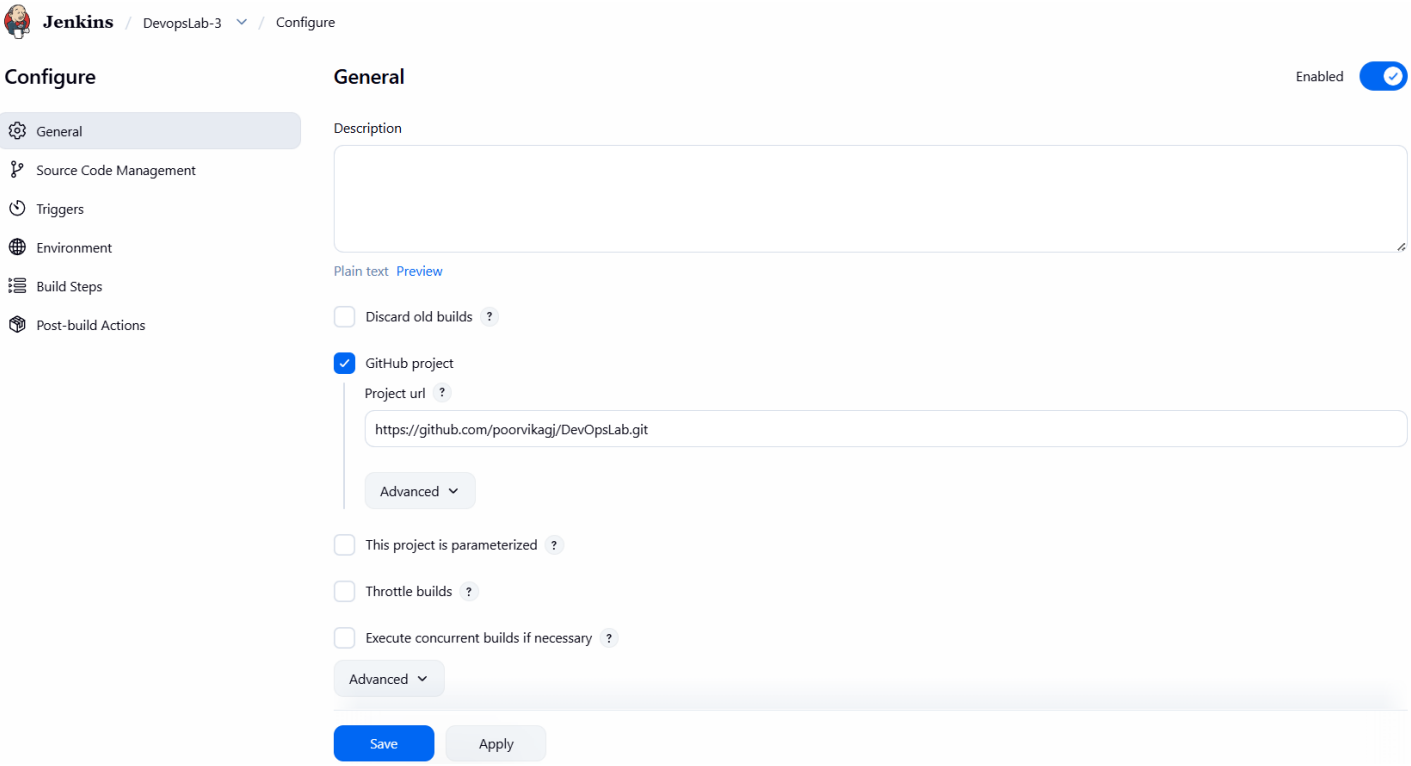


Using the freestyle Pipeline

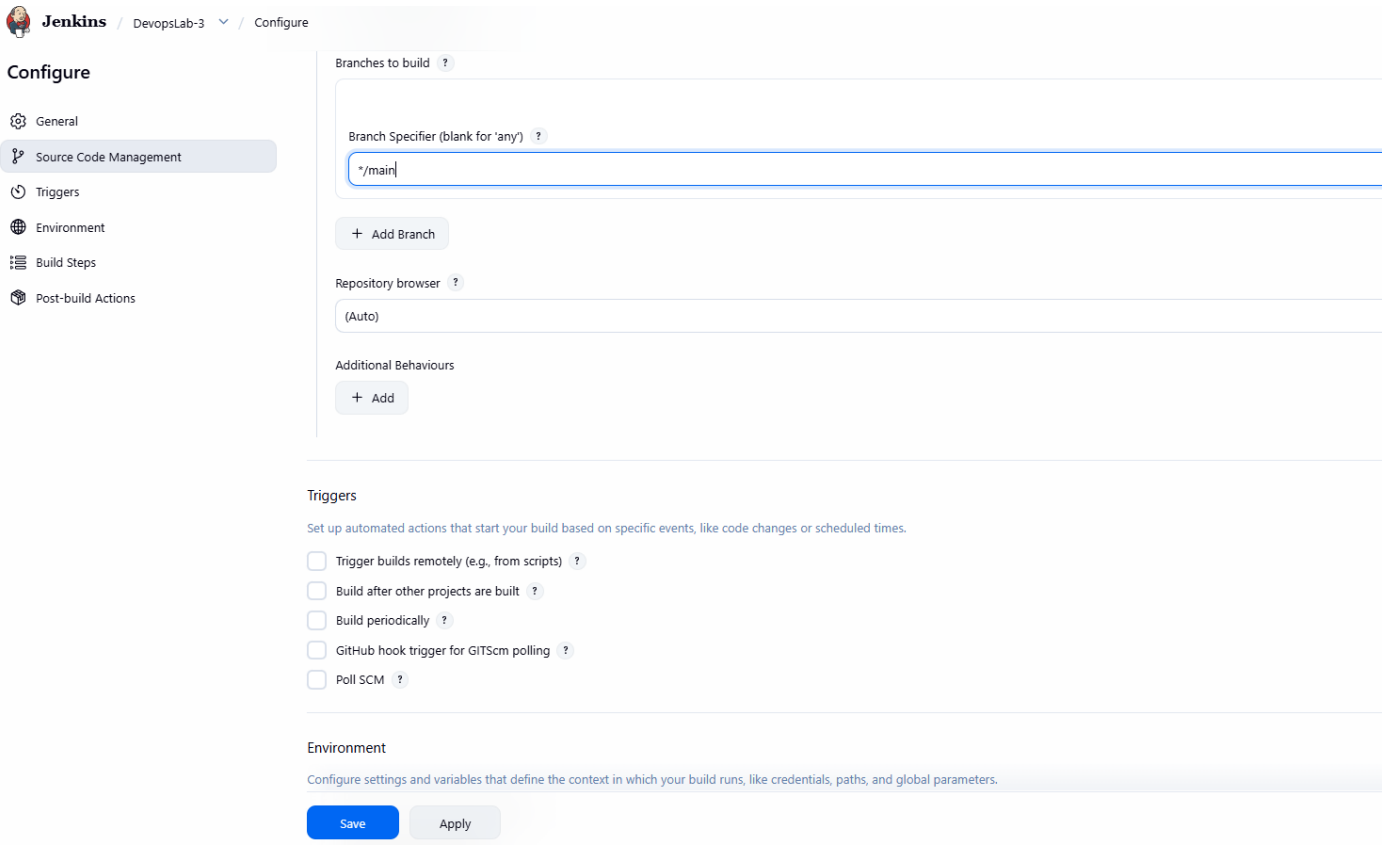
Select a new item to create a freestyle pipeline and name the pipeline.



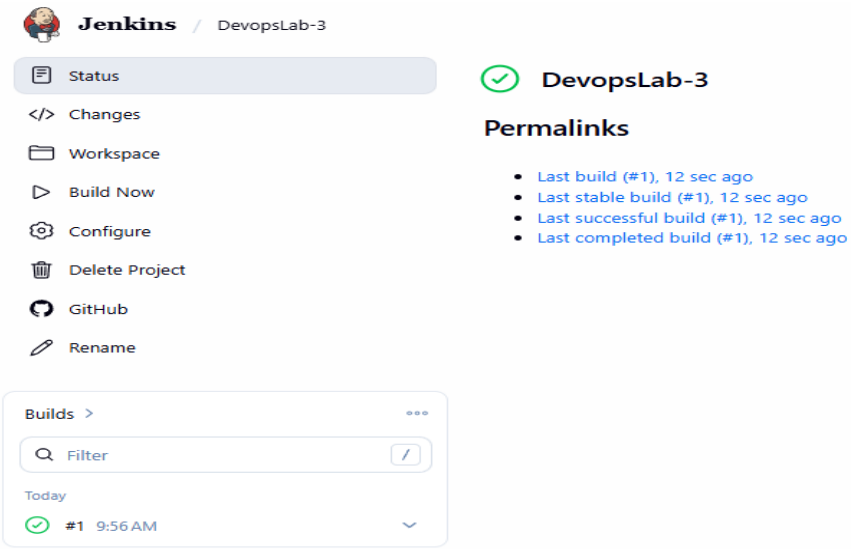
As the project is in the GitHub select GitHub project and paste the respective repo link.



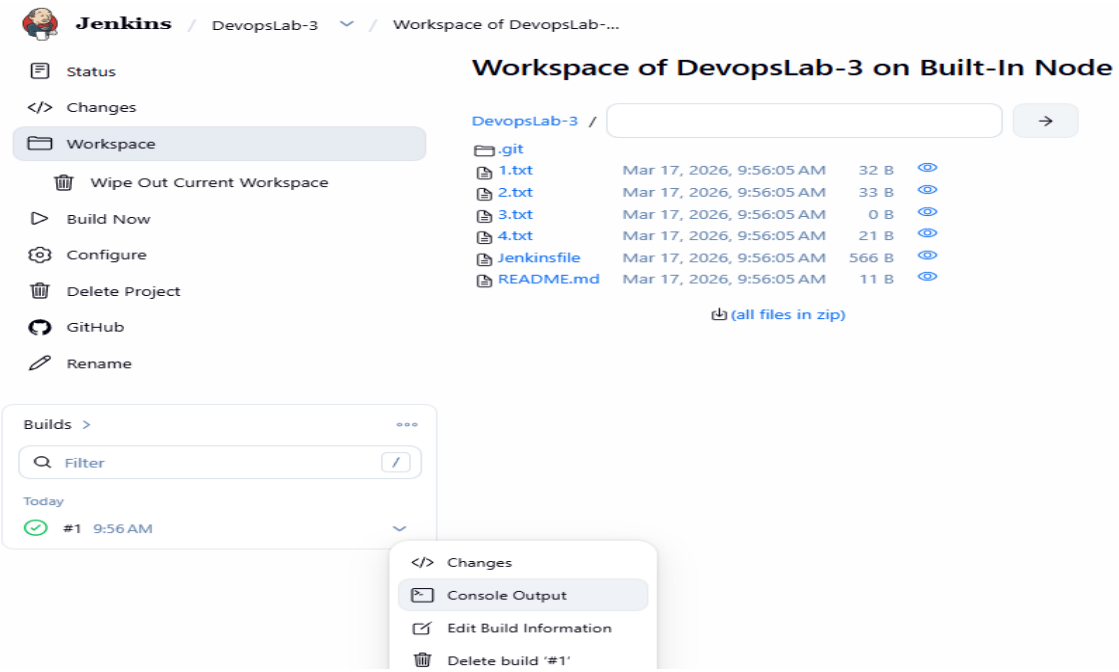
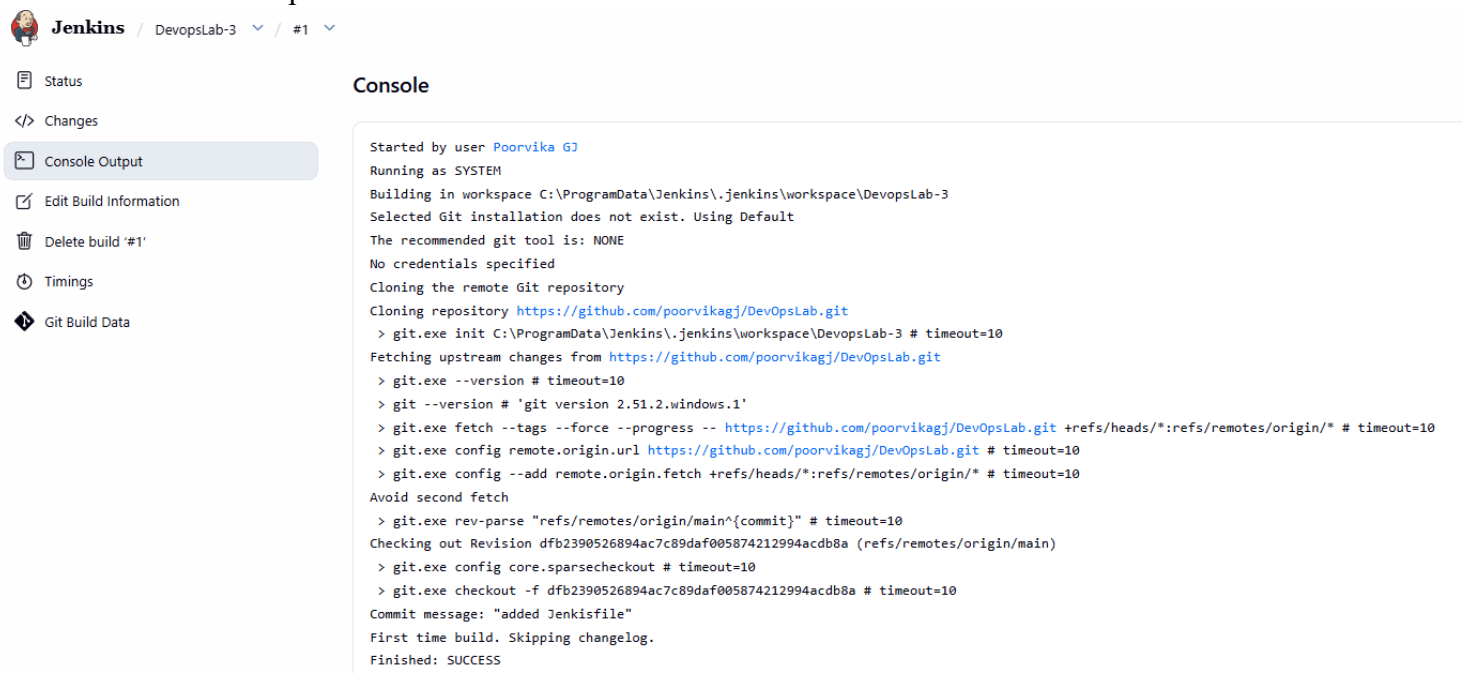
Write the branch name from where the pipeline should be built and save it and we choose the source code management as our Jenkinsfile is the project repo.



After the configuration, save the pipeline and build the pipeline.



Freestyle pipeline clones the repo into the workspace and you can see the files and to see the console ouput select the build up and see the output.



This is the Console output.